
Task #3 - Clustering (Unsupervised classification)

Cluster analysis is an exploratory data analysis technique aiming at getting insight into data. Adopting an informal definition, clustering can be stated as the problem of identifying natural or interesting groups in data. It is called unsupervised learning due to the lack of any information on cluster membership: no particular assignments of data items are available and usually the number of clusters is not known in advance.

The goal is intuitive, vaguely defined: given a data set, a partition of the data items is sought such that items belonging to the same cluster are similar while items belonging to different clusters are dissimilar. The work conducted on clustering converges only at this general level of description; as for concrete methods, there exist a wide range of clustering techniques based on different principles and yielding different results. Trying to unify the initial informal concept of clustering into an axiomatic framework governed by a unique objective function, Kleinberg (2002) obtains an impossibility result.

The definition of clustering leaves space to a wide choice of **objective functions** (to quantify the degree of "interestingness" or "naturalness" in groupings) and **similarity functions**, depending strongly on the domain under investigation. The choice is rarely straightforward.

Clustering is a difficult optimization problem: even if the number of clusters is fixed (the case of supervised clustering), the number of possible partitions increases exponentially with the number of objects; the size of the search space in this case is given by the Stirling number of the second kind. When the number of clusters is not known (the case of unsupervised clustering) the number of ways to partition a set of n objects into non-empty subsets is given by the n^{th} Bell number. For example, there are 2×10^{15} ways to partition a set of 25 objects into 5 groups, and more than 4×10^{18} ways to partition them when the number of clusters is not fixed.

Tasks Description

1. **The first task** is about *supervised* clustering - knowing the number of clusters. To get a practical understanding on the studied clustering algorithms, correlating the theoretical insights of each algorithm with its results in various contexts (data

distributions), each of the following algorithms must be executed on each of the data sets provided [here](#), considering as known the number of clusters:

- **kMeans**
- **EM** for clustering under Gaussian assumptions
- **hierarchical** algorithms under all known linkage metrics (single link, complete link, average link, Ward variance)
- **Density-based** algorithms (DBSCAN); experiment with different parameter settings to adapt to your data (**hyper-parameter tuning**)

Pay attention to **data standardization** (Z score normalization) - given that you use a Minkowski distance metric (alternatively try the Mahalanobis distance).

For each data set and each clustering algorithm, plot the clustering result (a two dimensional **scatterplot** of the data where data items are colored based on cluster assignments) and compute the **Adjusted Rand Index** - an external validation criteria that you use to compare the cluster assignments against true classes (it basically compares the agreement of two partitions). Make the analysis data-oriented (analyze each dataset in turn) such that you can conclude for each data set which was the most appropriate clustering technique and provide valid **justifications**.

For the **Iris data** set apply **Self Organizing Maps (SOM)** with at least 2 different map configurations, followed by kMeans applied on cell weights to extract 3 clusters. Visualize the distribution of the data on the map.

2. The second task is about *unsupervised* clustering - retrieving the natural number of clusters.

You will work with the **Iris** and **2d-10c** datasets for this task and two clustering algorithms: a **hierarchical** algorithm and **kMeans**.

You will experiment with two ways to identify the natural number of clusters:

- The elbow method
- An internal clustering evaluation index.

The **elbow** method plots a measure of cohesion of the clusters that depends on the clustering algorithm over the number of clusters in the partition; an elbow/knee is then identified. For kMeans the algorithm is applied several times with various numbers of clusters and the intra-class variance is plotted. For the hierarchical algorithm the dendrogram is cut at various levels and the values of the corresponding linkage metric

are plotted. Practically, few top values on the dendrogram are used (i.e. take the last x values in the vector giving the linkage values when joining clusters, and plot them). For the Iris data set the number of clusters should be varied from 2 to 10.

An internal clustering evaluation index is an unsupervised quality measure; unlike Adjusted Rand Index, it does not take into account class labels. Such an index is computed for different partitions with different numbers of clusters and is maximized/minimized. For this task use the **Silhouette Width** Index or the CritCF index (the last one aiming at joint unsupervised clustering and feature selection).

3. The third task aims at performing clustering analysis on the data set you chose for the classification task. Identify the natural number of clusters and check if the classes in data agree with natural groups by analyzing the contingency matrix and computing external validation criteria (ARI). Apply all the clustering algorithms again on this dataset (without SOM).

Resources

- DBSCAN - <http://www.aaai.org/Papers/KDD/1996/KDD96-037.pdf>
- HDBSCAN - <https://hdbscan.readthedocs.io/en/latest/>
- EM - https://www.cs.toronto.edu/~rgrosse/courses/csc411_f18/slides/lec16-slides.pdf
- SOM - http://www.pspc.unige.it/~drivsc/Papers/VanHulle_Springer.pdf
- Adjusted Rand Index (ARI) - https://en.wikipedia.org/wiki/Rand_index
- Silhouette Width - [https://en.wikipedia.org/wiki/Silhouette_\(clustering\)](https://en.wikipedia.org/wiki/Silhouette_(clustering))
- CritCF index - https://staff.fmi.uvt.ro/~daniela.zaharie/ma2016/projects/applications/feature_selection/featureSelection%2BClustering.pdf , implemented in R: <https://cran.r-project.org/web/packages/miclust/miclust.pdf>

Grading (See detailed score distributions in the grading excel spreadsheet)

Requirement	Points
T1 - The standard clustering algorithms for all data sets with plots, ARI and comments	5p
T1 - SOM on Iris	1p
T2 - Identifying the natural number of clusters - elbow+index	2p
T3 - Cluster analysis on the dataset used for the classification task	2p